

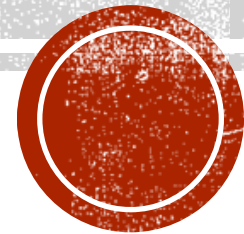
SMART CUSTOMER SERVICE REQUEST MANAGEMENT USING QUEUE ADT

CSA0316-DATA STRUCTURES

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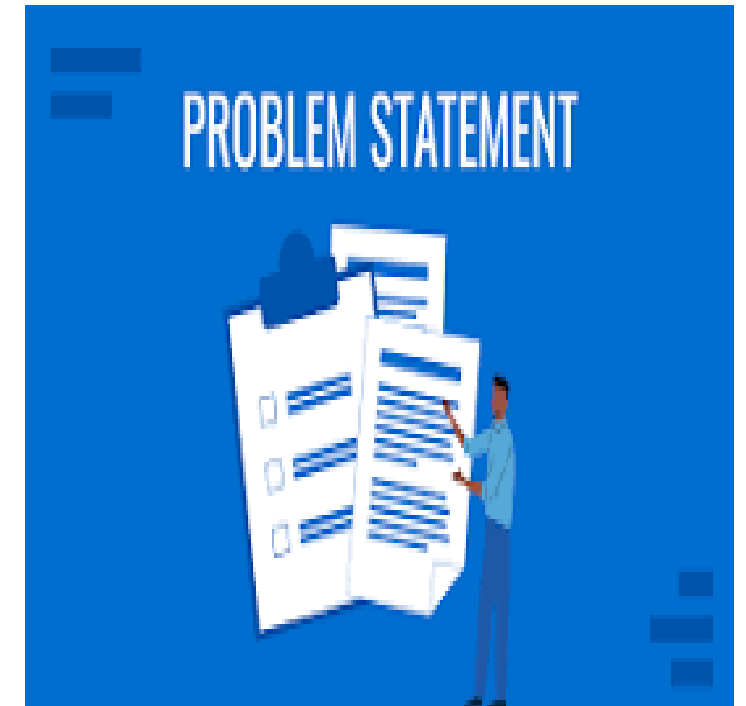
INTRODUCTION

- Customer service request management helps handle customer queries and complaints.
- It keeps all customer requests in an organized order.
- The **Queue ADT** follows the **First In, First Out (FIFO)** principle.
- The customer who sends the request first is served first.
- This makes the service fair, fast, and efficient.



PROBLEM STATEMENT

- Customer service centers receive a large number of requests every day.
- Without a proper system, requests may not be processed in the order they are received.
- Delays and missed requests can lead to customer dissatisfaction.
- Manual request handling becomes difficult as the number of customers increases.



ABSTRACT DATA TYPES (ADT) OVERVIEW

- An **Abstract Data Type (ADT)** is a logical model that defines how data is stored and the operations that can be performed on it.
- It focuses on **what the data structure does**, not **how it is implemented**.

Common ADTs

- **Stack** – Follows **Last In First Out (LIFO)**.
- **Queue** – Follows **First In First Out (FIFO)**.
- **Linked List** – Stores data as connected nodes for flexible insertion and deletion.
- **Tree** – Organizes data in a hierarchical structure for efficient searching and management.



QUEUE ADT

- A **Queue** is a linear Abstract Data Type (ADT) that stores elements in a sequential order.
- It follows the **First In, First Out (FIFO)** principle, where the first element added is the first one removed

Queue Operations

- **Enqueue** – Adds a new request to the rear of the queue.
- **Dequeue** – Removes the request from the front of the queue.
- **Peek (Front)** – Displays the first request without removing it.



Queue Data Structure



WHY QUEUE IS SUITABLE FOR CUSTOMER SERVICE

- **Fair Request Processing:** Requests are handled in the same order they are received (FIFO).
- **No Request is Skipped:** Every customer request is processed one by one without being missed.
- **Efficient Order Management:** Keeps all requests organized and easy to manage.
- **Reduced Waiting-Time Conflicts:** Customers know their requests will be handled in sequence.
- **Improved Customer Satisfaction:** Fair and systematic processing builds trust and improves service quality.

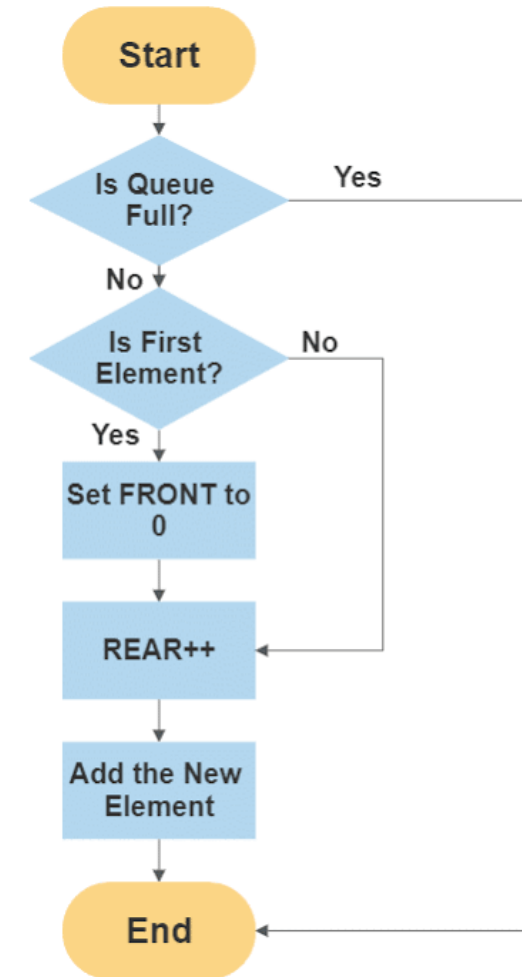


WORK FLOW EXAMPLE

Customer Request Flow

- Customers submit their service requests.
- Each request is added to the **rear** of the queue.
- The support agent processes the request at the **front** of the queue.
- After completion, the next request is processed.
- **Process step by step:**

Step	Queue Status
Customer A submits a request	A
Customer B submits a request	A → B
Customer C submits a request	A → B → C
Agent completes Customer A's request	B → C
Agent completes Customer B's request	C

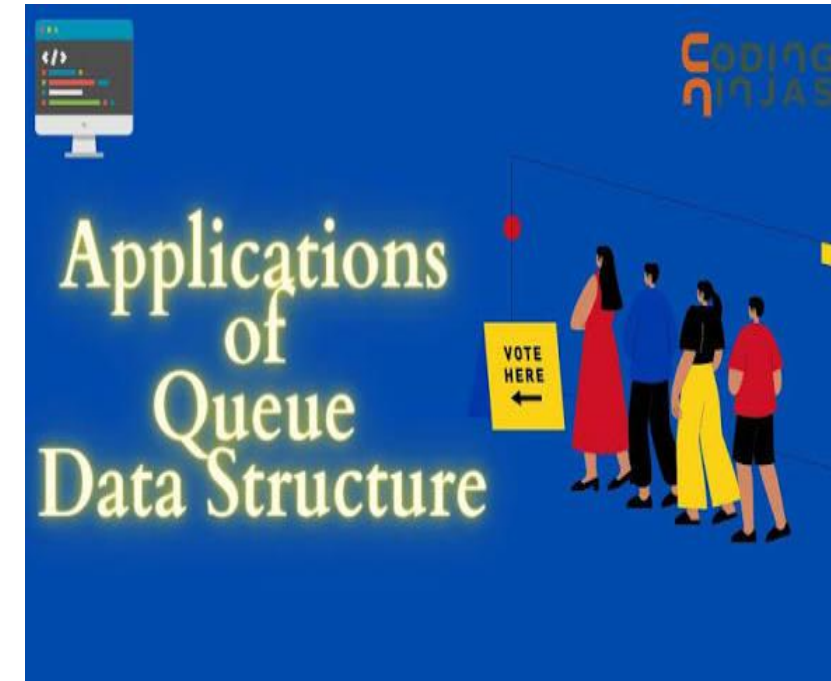


ENQUEUE



REAL-WORLD APPLICATIONS

- **Call Centers:** Incoming customer calls are answered in the order they are received.
- **Bank Token Systems:** Customers are served based on their token number (first come, first served). .
- **Hospital Registration:** Patients are registered and attended in sequence.
- **Online Customer Support:** Chat and email requests are handled using a queue to ensure fair service.



CONCLUSION

Benefits of Queue ADT

- Ensures fair processing using the **FIFO** principle.
- Organizes customer requests efficiently.
- Reduces waiting time and prevents skipped requests.
- Improves customer satisfaction and service quality.

Summary

- Queue ADT is the most suitable data structure for customer service request management.
- It processes requests in the order they are received, making the system simple, fair, and reliable.



Thank You!



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